

Solution Requirements

Date	29 September 2026
Team ID	SWTID-2026-1374
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The requirements below define what the solution will do and how well it will perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Test users are created in ServiceNow. Users log in with their own credentials so their identity and assigned roles can be verified before any record is accessed.	High
2	Authorization levels	Custom roles are created and assigned to users: bb1 – Read, bb2 – Create, bb3 – Write/Edit, bb4 – Delete. Admin has full access to all records.	High
3	External interfaces	No third-party API or external database is required. The solution works fully inside the ServiceNow platform.	Low
4	Transactions processing	The Institution Details table stores student information. Read, Create, Write and Delete operations on records are each controlled by their own ACL.	High
5	Reporting	List views of the Institution Details table show only the records a user is permitted to see. Administrators can view records of all branches (ECE, EEE, CSE).	Medium
6	Business rules, etc.	The READ ACL uses a script and a Branch-based condition to restrict EEE branch records to authorized roles. Administrators access all records and normal users are restricted by their assigned roles.	High
7	Compliance to laws or regulations	Student data must be kept confidential and accessible only to authorized users, following the institution's data privacy and least-privilege access policies.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
8	Other	Records are created for ECE, EEE and CSE branches to test the ACLs, and access is verified for each role by logging in or impersonating the test users.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	ACL checks should not delay opening or listing records.	List and form pages load within 3 seconds.
2	Scalability	New roles, branches or users can be added without redesigning the ACLs.	A new branch or role is added by changing only the ACL condition/script.
3	Security & Data Privacy	Only authorized roles can read, create, edit or delete records; EEE records are hidden from unauthorized users.	100% of unauthorized read, create, write and delete attempts are denied.
4	Reliability & Availability	ACLs are enforced consistently on every request on the ServiceNow platform.	Same access result for the same role on every test.
5	Usability & Accessibility	Administration is easy: roles are assigned from the user record using standard ServiceNow forms.	A role is assigned to a user in a few steps using the standard UI.
6	Other	Data confidentiality and controlled record access based on role and Branch value.	Each role sees only the records and operations allowed for it.